$x=1$ $\quad P(X\leq 1) = P(x=0)+P(x=1)$

$$= 4C_0\left(\frac{3}{5}\right)^0\left(\frac{2}{5}\right)^4 + 4C_1\left(\frac{3}{5}\right)^1\left(\frac{2}{5}\right)^3$$

$$= \frac{16}{625} + \frac{96}{625} = \frac{112}{625} < \frac{1}{2}$$

$x=2$ $\quad P(X\leq 2) = P(x=0)+P(x=1)+P(x=2)$

$$= \frac{112}{625} + P(x=2) = \frac{112}{625} + 4C_2\left(\frac{3}{5}\right)^2\left(\frac{2}{5}\right)^2$$

$$= \frac{112}{625} + \frac{216}{625} = \frac{328}{625} > \frac{1}{2}$$

Therefore the sample should contain a minimum of 2 boys.

10.1.3 POISSION DISTRIBUTION

Introduction

In a Binomial distribution with parameter n and p if the exact value of n is not definitely known and if p is very small then it is not possible to the find the binomial probabilities .Even if n is known and it is very large, calculations are tedious. In such situations a distribution called Poisson distribution is very much useful.

In 1837 French mathematician Simeon Dennis Poisson derived the distribution as a limiting case of Binomial distribution. It is called after his name as Poisson distribution.

Conditions:

(i) The number of trails 'n' is indefinitely large i.e., $n \to \infty$

(ii) The probability of a success 'p' for each trial is very small i.e., $p \to 0$

(iii) $np = \lambda$ is finite

(iv) Events are Independent

Definition

A random variable X is said to follow a Poisson distribution if it assumes only non-negative integral values and its probability mass function is given by

$$P(X=x) = \begin{cases} \dfrac{e^{-\lambda}\lambda^x}{x!}; & x = 0,1,2,..... \\ 0 & otherwise \end{cases}$$